

# Optics Letters

## Low-noise 750 MHz spaced ytterbium fiber frequency combs

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Received 8 June 2018; revised 11 July 2018; accepted 13 July 2018; posted 16 July 2018 (Doc. ID 334691); published 22 August 2018

**We demonstrate two low-noise 750 MHz ytterbium fiber frequency combs that are independently stabilized to a continuous-wave laser. A bulk electro-optic modulator and a single-stack piezo-electric transducer are employed as fast actuators for stabilizing the respective cavity length to heterodyne beat notes. Both combs exhibit in-loop fractional frequency instabilities of  $\sim 10^{-18}$  at 1 s. To the best of our knowledge, this is the first demonstration of tightly phase-locked ( $<1$  rad root mean square phase noise integrated from 0.1 Hz to 10 MHz) fiber frequency combs with 750 MHz fundamental repetition rate. © 2018 Optical Society of America**

**OCIS codes:** (320.7090) Ultrafast lasers; (140.3510) Lasers, fiber; (120.3930) Metrological instrumentation.

<https://doi.org/10.1364/OL.43.004136>

The development of optical frequency combs (OFCs) over the past decade has experienced fascinating evolution toward high-frequency stability, high coherence, high repetition rate ( $f_{\text{rep}}$ ), and high output power. This is particularly evident when we look at the wide variety of OFCs-based numerous exciting applications, such as ultra-low noise microwave generation [1,2], optical waveform synthesizers [3,4], astronomical calibration [5], and dual-comb spectroscopy [6]. Elevating the repetition rate to  $\sim$ GHz levels on the premise of low noise will have further benefits in increasing the scan speed used in dual-comb distance measurements [7], improving the sensitivity of dual-comb spectroscopy [8], and extending the measurable range of one-shot 3D shape imaging [9]. Multiple configurations of frequency combs with  $\sim$ GHz repetition rates, including solid-state lasers, fiber lasers, and micro-resonators, have been reported [10–15]. High-repetition-rate ytterbium (Yb) frequency combs are prominent, owing to their exceptional characteristics such as superior power scalability and feasible

Si-photo detector measurements. Phase locking the comb to an optical frequency reference other than a radio frequency (RF) standard can reduce the phase error to  $<1$  rad and narrow the linewidth to sub-mHz levels [16]. This is propitious to increase frequency stability by four to six orders of magnitude in, e.g., an 800 MHz Ti:sapphire laser [17]. However, similar phase-locking records in fiber-laser-based frequency combs have been obtained only at a relatively low repetition rate (typically around 100 MHz) [16,18,19].

The tight phase locking in high-repetition-rate Yb: fiber frequency combs involves two issues. One is the insufficient servo bandwidth against high intrinsic noise, and the other is cross-talk between the intra-cavity electro-optic modulator (EOM) used to stabilize the cavity length and gain modulation to the carrier-envelope frequency ( $f_{\text{ceo}}$ ).

Larger servo bandwidths are required because the intrinsic noise, linewidth, or timing jitter generally scales up with the repetition rate [10,11]. Therefore, it is necessary to achieve  $\sim$ MHz-level servo bandwidth for  $f_{\text{ceo}}$  with  $>100$  kHz free-running linewidth, which is a common characteristic of Yb: fiber lasers with high repetition rate. Graphene modulators are capable of producing large loss modulation bandwidths and have been used to successfully lock  $f_{\text{ceo}}$  with  $\sim 1$  MHz servo bandwidth in a thulium fiber comb [20] and a servo bandwidth of 1.6 MHz in an erbium fiber comb [19], respectively. However, the graphene modulator is not suitable to be applied to the  $\sim 1$   $\mu\text{m}$  wavelength range, as the doping concentration would need to be too much higher than in 1.5  $\mu\text{m}$  or 2  $\mu\text{m}$  wavelength to achieve the desired transparency. On the other hand, it was also reported that gain modulation bandwidth of more than a few hundred kHz has been achieved in an Er: fiber comb by directly feeding back the modulated current to the pins of the pump laser diode (LD) [18]. However, only a few publications in the literature report Yb: fiber combs with tight phase locking of  $f_{\text{ceo}}$  [16,21]. Those reported combs benefit from the record free-running narrow linewidth ( $<10$  kHz) of  $f_{\text{ceo}}$  for fiber lasers.

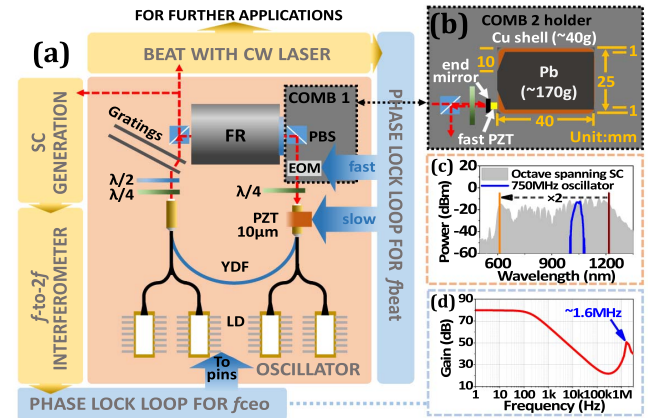
Cross-influence occurs when an intra-cavity EOM is involved. Minute misalignment of the EOM crystal causes coupling between cavity length modulation and pulse polarization via the complex laser dynamics [22], hence introducing competition between the stabilizations of  $f_{\text{ceo}}$  and  $f_{\text{beat}}$ . An alternative way to overcome this drawback is to use a piezo actuator that can potentially operate with a response bandwidth of a few hundred kHz [23]. Piezo-electric transducers (PZTs) can modulate the cavity length more purely than EOMs, since the PZTs barely influence the intra-cavity dispersion and quality factor, and are less dependent on the polarization and wavelength.

We presented a fully stabilized 750 MHz Yb: fiber comb phase locked to both microwave standards and optical references in a previous paper [24]. However, the residual phase errors of  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  were still high (13.7 rad and 4.6 rad, respectively), and the in-loop instability was  $10^{-15}$  at 1 s. Recently, Endo *et al.* reported a 1 GHz monolithic solid-state-laser-based OFC with  $7 \times 10^{-19}$  at 1 s fractional frequency instability [12]. A  $f_{\text{ceo}}$ -stabilized (4.5 rad root mean square phase) 1.8 GHz semiconductor disk laser frequency comb and a fully stabilized 1.05 GHz diode-pumped solid-state laser frequency comb with  $10^{-12}$  at 1 s relative frequency instability were also demonstrated [10,11]. For fiber-based  $\sim$ GHz frequency combs, a counter limited  $4 \times 10^{-11}$  at 1 s frequency instability was achieved in a 1.04 GHz Yb: fiber laser fully stabilized to an RF reference [14]. In this work, we demonstrate phase-stabilized 750 MHz Yb: fiber frequency combs with  $<1$  rad root mean square phase both in  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  in two distinct configurations. Both combs exhibit in-loop fractional frequency instability of  $10^{-18}$  at 1 s. To the best of our knowledge, this is the first report of tightly phase-locked fiber-based combs with the highest fundamental repetition rate of 750 MHz.

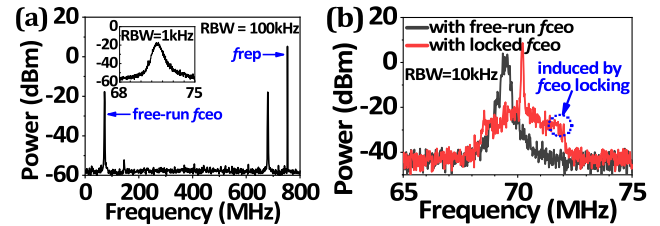
We marked an Yb: fiber comb equipped with an intra-cavity EOM (a 2 mm long, z-cut LiNbO<sub>3</sub> crystal) in transmission configuration as comb 1, and the other comb with a single-stack PZT (PL022.3x, Physik Instrumente GmbH) in reflection configuration as comb 2. Slow PZTs with 10  $\mu$ m travel range are separately mounted in both combs, which are responsible for the slow drifts below 2 kHz.

Comb 1 is based on the ring-cavity 750 MHz Yb: fiber laser described in Ref. [24], and its cavity configuration is shown in Fig. 1(a). Nonlinear polarization rotation (NPR) was employed to initiate the mode locking under a small negative (around  $-0.001$  ps<sup>2</sup>) net dispersion. The full-width-half-maximum (FWHM) of the optical spectrum [see Fig. 1(c)] was 27 nm, and the laser delivered a train of near-transform-limited pulses with a  $\sim 70$  fs pulse duration. The major part of the oscillator output power ( $\sim 630$  mW) was directly launched into a homemade tapered photonic crystal fiber (PCF) for generating an octave spanning super-continuum (SC) spectrum [Fig. 1(c)]. The total length of the fabricated tapered PCF was  $\sim 25$  cm long, including 5 cm tapered length with 1.8  $\mu$ m waist core diameter, two symmetric 3 cm long transition parts and  $\sim 14$  cm long untapered input/output section with 4.5  $\mu$ m original core diameter. In order to generate one octave spanning spectrum, the pitch of the commercial PCF was decreased to ensure the dual zero dispersion wavelengths (800 nm and 1040 nm, respectively).

A free-running  $f_{\text{ceo}}$  signal resolution bandwidth [RBW = 100 kHz, plotted in Fig. 2(a)] with  $>40$  dB signal-to-noise ratio (SNR) was obtained using a conventional f-to-2f



**Fig. 1.** (a) Configuration of comb 1. (PBS: polarization beam splitter, YDF: ytterbium-doped fiber.) (b) Structure of the fast PZT section in comb 2. (c) Octave-spanning SC generation (blue curve: optical spectrum of the 750 MHz oscillator). (d) Open-loop response of the PID circuit for locking  $f_{\text{ceo}}$ .



**Fig. 2.** (a) RF spectrum of free-running  $f_{\text{ceo}}$  signal. Inset shows a magnified view. (b) Free-running  $f_{\text{beat}}$  signals with free-running (gray curve) and locked (red curve)  $f_{\text{ceo}}$ .

interferometer. A small portion ( $\sim 5$  mW) of the output power was used to beat with the 1064 nm continuous-wave (CW) laser (a monolithic isolated single-mode end-pumped ring laser with approximately 3 kHz free-running linewidth). In the servo loops, the generated  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  signals were individually filtered, electronically amplified, and directly phase locked to the 70 MHz frequency references (Keysight 33600A) using digital phase discriminators and homemade proportional-integral-differential (PID) loop filters. Several considerations are given in order to realize phase locking with  $<1$  rad root mean square phase in the 750 MHz Yb: fiber combs.

First, an optimum mode-locking status in the stretched pulse regime with an appropriate LD bias current is an indispensable precondition. Only in this way can it be possible to acquire a maximized linear  $f_{\text{ceo}}$  tuning rate ( $\Delta f_{\text{ceo}}/\Delta I_{\text{LD}} = 2.1$  MHz/mA in this experiment) while simultaneously maintaining invariable free-running linewidth ( $\sim 200$  kHz) and SNR within a large current tuning range ( $\sim 800$  mA). This is a key factor, since we found it was difficult to observe a coherent peak in the locked  $f_{\text{ceo}}$  signal if the oscillator operated at a low  $f_{\text{ceo}}$  tuning rate (e.g.,  $<0.3$  MHz/mA).

Second, employing digital phase discriminators instead of frequency dividers favors authentically revealing the unaltered features of the beat signals. Due to the relatively high intrinsic noise of  $f_{\text{ceo}}$  and  $f_{\text{beat}}$ , analog mixers are not adequate to detect such broad and rapidly jittery signals without the help of

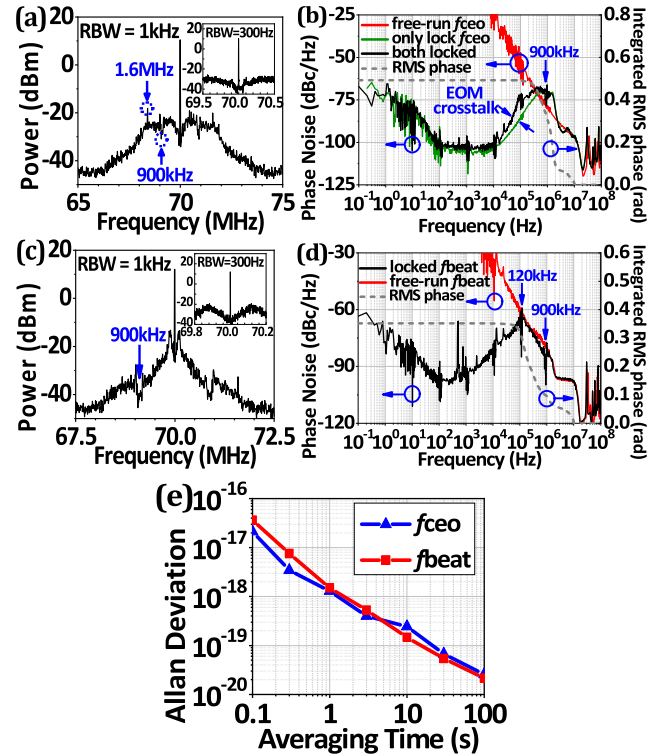
frequency dividers. In order to prevent the degradation of frequency stability induced by frequency divisions, digital phase discriminators (ON Semiconductor MCH12140) with 800 MHz working bandwidth are employed. Owing to the available phase detection range being  $-2\pi$  to  $+2\pi$ , it is able to detect such large phase fluctuations without a frequency divider, and high-speed servo control is applicable.

Third, two homemade, high-speed PID loop filters must be specifically tailored to the high-repetition-rate Yb: fiber frequency combs for the locking of  $f_{\text{ceo}}$  and  $f_{\text{beat}}$ . For the  $f_{\text{ceo}}$  loop filter, two differentiator stages were employed to compensate for the decreasing current modulation response at high frequency. The increasingly high open-loop gain was cutoff at  $\sim 1.6$  MHz [as shown in Fig. 1(d)], which is limited by the gain-bandwidth product (GBP) of the specific operational amplifiers. With respect to  $f_{\text{beat}}$ , the voltage applied to the EOM ranged within  $0\text{ V} \pm 1\text{ V}$ , thus cooperating with the slow PZT for cavity length stabilization. The outputs of the PID circuits directly feed back to the LD electrodes and the EOM electrodes to maintain the desired bandwidth.

The RF spectra and phase noise spectra of simultaneously phase-locked  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  of comb 1 are shown in Fig. 3. The insets in Figs. 3(a) and 3(c) show the RBW-limited linewidth for both signals, indicating coherent peaks with 50 dB (RBW = 300 Hz) SNR, which are equivalent to  $\sim 75$  dB  $\times$  Hz ( $\sim 75$  dB in 1 Hz RBW). Obvious dents at  $\sim 900$  kHz in both Figs. 3(a) and 3(c) are induced by the strong EOM resonance applied to both  $f_{\text{ceo}}$  and  $f_{\text{beat}}$ . Figures 3(b) and 3(d) show considerable phase noise reductions in both signals after stabilization. The residual root mean square phase errors are, respectively, calculated to be 0.49 rad for  $f_{\text{ceo}}$  and 0.35 rad for  $f_{\text{beat}}$  with 0.1 Hz to 10 MHz integration range. This indicates energy concentrations of 81% and 89% in the corresponding coherent peaks, respectively. Both  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  exhibit in-loop fractional frequency instability of  $\sim 1.5 \times 10^{-18}$  with an averaging time of 1 s and a frequency instability of  $\sim 2.5 \times 10^{-20}$  at 100 s averaging time [counted by Agilent 53132 A and normalized to 282 THz, as shown in Fig. 3(e)]. All the RF frequency synthesizers, frequency counters, and phase noise analyzers are referenced to a GPS-disciplined rubidium (Rb) oscillator. One should note that there is a distinct rising tendency when the EOM was enabled [see Fig. 3(b)]. This is caused by crosstalk between the EOM and  $f_{\text{ceo}}$ . Since crosstalk can severely deteriorate the locked  $f_{\text{ceo}}$  if the EOM bandwidth is fully exploited, a tradeoff must be made by keeping the cutoff frequency of  $f_{\text{beat}}$  servo loop at  $\sim 120$  kHz, although the resonance frequency of EOM is  $\sim 900$  kHz.

In order to reduce crosstalk originating from the intra-cavity components, we built comb 2, which employs a high-speed single-stack PZT (fast PZT) for stabilizing the cavity length. Accordingly, the cavity must be modified as “ $\sigma$ ” type, so that the laser can have a linear path for the implementation of the end mirror driven by the fast PZT. The configuration of comb 2 is schematically illustrated by simply replacing the gray highlighted area in Fig. 1(a) with Fig. 1(b) and leaving the remaining structures unchanged. It is believed that such a modification scarcely influences the optical characteristics of the laser oscillator; thus, both the free-running  $f_{\text{ceo}}$  and  $f_{\text{beat}}$  signals have nearly the same behavior as in comb 1.

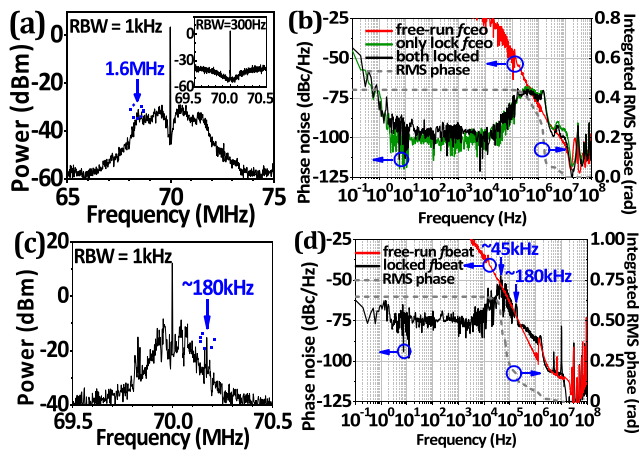
To exploit the highest response bandwidth of the fast PZT, we engineered a special holder following the technique



**Fig. 3.** In-loop phase-locking results for comb 1: (a) in-loop  $f_{\text{ceo}}$  with simultaneously stabilized  $f_{\text{beat}}$ . The inset shows a magnified view with smaller span and RBW. (b) Phase noise spectra of  $f_{\text{ceo}}$  (free-running  $f_{\text{ceo}}$ : red solid; phase locked  $f_{\text{ceo}}$  with free-running  $f_{\text{beat}}$ : green solid; and simultaneously stabilized  $f_{\text{beat}}$ : black solid) and integrated phase error of  $f_{\text{ceo}}$  under full phase locking (gray dashed). (c) In-loop  $f_{\text{beat}}$  with simultaneously stabilized  $f_{\text{ceo}}$ . The inset shows a magnified view. (d) Phase noise spectra of  $f_{\text{beat}}$  (fully free-running: red solid; full stabilization: black solid) and the integrated phase error of  $f_{\text{beat}}$  under full phase locking (gray dashed curve). (e) Relative Allan deviations (normalized to 282 THz) of the in-loop  $f_{\text{ceo}}$  (blue triangle) and  $f_{\text{beat}}$  (red square) under full stabilization.

introduced in Ref. [23]. As shown in Fig. 1(b), the PZT holder is a hollow copper (Cu) cylinder filled with lead (Pb) to dampen longitudinal resonance waves. One end inside the cylinder was cone-shaped to prevent acoustic resonance. A  $3 \times 3 \times 1$  mm<sup>3</sup> tiny end mirror and a single-stack PZT were bonded to the edge of the holder with a rigid adhesive (CrystalBond 590). When adhering the PZT, we melted the adhesive and applied  $\sim 20$  MPa pressure to ensure no bubbles and increase the spring constant. Moreover, low-order drum modes in the copper could be avoided by placing the PZT off-center. All those procedures are required to keep the lowest resonance frequency of the PZT to  $\sim 180$  kHz. This damping cylinder was stuck by an annular holder mounted on the baseplate.

The phase-locking results of comb 2 are shown in Fig. 4. Of great importance, the stabilized  $f_{\text{ceo}}$  spectrum was no longer affected by the  $f_{\text{beat}}$  locking loop [Fig. 4(b)]; therefore,  $f_{\text{ceo}}$  exhibits a smoother RF spectrum and has a lower integrated residual phase error than comb 1 [Fig. 4(a)]. Two peaks at  $\sim 180$  kHz were observed, as shown in Fig. 4(c). Those peaks are identified as the resonance of the fast PZT. To avoid this, the effective servo bandwidth was bound to  $\sim 45$  kHz [see Fig. 4(d)]. The calculated



**Fig. 4.** In-loop phase-locking results for comb 2: (a) RF spectra of  $f_{cco}$ . Inset shows a magnified view. (b) Phase noise spectra of  $f_{cco}$  (free-running: red solid; only locked  $f_{cco}$ : green solid; full phase locking: black solid) and integrated phase error of  $f_{cco}$  (gray dashed). (c) RF spectrum of in-loop  $f_{beat}$ . (d) Phase noise spectra of  $f_{beat}$  (free-running: red solid; full phase locking: black solid) and the integrated phase error of  $f_{beat}$  (gray dashed curve).

root mean square phase errors are 0.44 rad for  $f_{cco}$  and 0.65 rad for  $f_{beat}$  (corresponding to 84% and 70% energy concentrations), respectively. Both in-loop frequency Allan deviations of comb 2 are at the same level as comb 1.

The overall experimental results show that both the EOM and PZT are able to achieve the tight phase locking in high-repetition-rate fiber combs with the same level of frequency stability. An EOM suppresses noise with a larger servo bandwidth, but it is inhibited by competition with  $f_{cco}$ . On the other hand, the PZT does not interfere with  $f_{cco}$ , but rather exhibits a relatively smaller servo bandwidth limited by the mechanics. It should be emphasized that good locking performance of  $f_{cco}$  plays an important role for achieving tight phase locking of  $f_{beat}$ . This is because stabilization of  $f_{cco}$  remarkably narrowed the free-running linewidth of  $f_{beat}$  [see Fig. 2(b)]. The consequent benefit is the reduced bandwidth requirements of the servo loop for both the EOM and PZT. Consequently, even a response bandwidth of 180 kHz in the PZT is sufficient to phase lock  $f_{beat}$ .

In conclusion, we demonstrated two tightly phase-locked 750-MHz Yb: fiber combs stabilized to an optical reference with low residual phase errors. The  $f_{cco}$  with  $\sim 200$  kHz free-running linewidth was phase locked to exhibit a  $\sim 75$  dB  $\times$  Hz coherent peak using simple LD current modulation. Furthermore, an EOM and PZT were exploited for tight phase locking of  $f_{beat}$  in such high-repetition-rate Yb: fiber combs. Both are able to suppress the root mean square phase to sub-radian level. The in-loop fractional frequency instability with either modulator can reach  $\sim 1.5 \times 10^{-18}$  at 1 s. The experiment results prove that both are practical schemes for low-noise operation of the frequency combs. The aforementioned techniques are simple and powerful tools for achieving low noise with high repetition rate in Yb: fiber combs for advanced applications. In addition, one of the combs is also amplified to 10 W with sub-100 fs de-chirped pulse duration, and the noise properties will be investigated and reported elsewhere.

**Funding.** Exploratory Research for Advanced Technology (ERATO), Japanese Science and Technology Agency (MINOSHIMA Intelligent Optical Synthesizer Project, JPMJER1304); National Natural Science Foundation of China (NSFC) (61575004, 61735001).

**Acknowledgment.** The authors are in debt to Hajime Inaba of the National Institute of Advanced Industrial Science and Technology (AIST), Kenichi Nakagawa of the University of Electro-Communications (UEC), and Haochen Tian of Tianjin University (TJU) for their generous suggestions.

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## REFERENCES

- J. Millo, M. Abgrall, M. Lours, E. M. L. English, H. Jiang, J. Guéna, A. Clairon, S. Bize, Y. Le Coq, and G. Santarelli, *Appl. Phys. Lett.* **94**, 141105 (2009).
- T. M. Fortier, M. S. Kirchner, F. Quinlan, J. Taylor, J. C. Bergquist, T. Rosenband, N. Lemke, A. Ludlow, Y. Jiang, C. W. Oates, and S. A. Diddams, *Nat. Photonics* **5**, 425 (2011).
- S. Chia, G. Cirmi, S. Fang, G. M. Rossi, O. D. Mücke, and F. X. Kärtner, *Optica* **1**, 315 (2014).
- Z. Jiang, C. Huang, D. E. Leaird, and A. M. Weiner, *Nat. Photonics* **1**, 463 (2007).
- C. Li, A. J. Benedick, P. Fendel, A. G. Glenday, F. X. Kärtner, D. F. Phillips, D. Sasselo, A. Szentgyorgyi, and R. L. Walsworth, *Nature* **452**, 610 (2008).
- I. Coddington, N. R. Newbury, and W. C. Swann, *Optica* **3**, 414 (2016).
- I. Coddington, W. C. Swann, L. Nenadovic, and N. R. Newbury, *Nat. Photonics* **3**, 351 (2009).
- A. Nishiyama, S. Yoshida, T. Hariki, Y. Nakajima, and K. Minoshima, *Opt. Express* **25**, 31730 (2017).
- T. Kato, M. Uchida, and K. Minoshima, *Sci. Rep.* **7**, 3670 (2017).
- S. Hakobyan, V. J. Wittwer, P. Brochard, K. Gürel, S. Schilt, A. S. Mayer, U. Keller, and T. Südmeyer, *Opt. Express* **25**, 20437 (2017).
- N. Jornod, K. Gürel, V. J. Wittwer, P. Brochard, S. Hakobyan, S. Schilt, D. Waldburger, U. Keller, and T. Südmeyer, *Optica* **4**, 1482 (2017).
- M. Endo, T. D. Shoji, and T. R. Schibli, *IEEE J. Sel. Top. Quantum* **24**, 1 (2018).
- P. Del'Haye, O. Arcizet, A. Schliesser, R. Holzwarth, and T. J. Kippenberg, *Phys. Rev. Lett.* **101**, 053903 (2008).
- I. Hartl, H. A. McKay, R. Thapa, B. K. Thomas, A. Ruehl, L. Dong, and M. E. Fermann, *Advanced Solid-State Photonics* (2009), paper MF9.
- D. Chao, M. Y. Sander, G. Chang, J. L. Morse, J. A. Cox, G. S. Petrich, L. A. Kolodziejski, F. X. Kärtner, and E. P. Ippen, *Optical Fiber Communication Conference* (2012), paper OW1C.2.
- T. R. Schibli, I. Hartl, D. C. Yost, M. J. Martin, A. Marcinkiewicz, M. E. Fermann, and J. Ye, *Nat. Photonics* **2**, 355 (2008).
- S. Fang, H. Chen, T. Wang, Y. Jiang, Z. Bi, and L. Ma, *Appl. Phys. Lett.* **102**, 231118 (2013).
- Y. Nakajima, H. Inaba, K. Hosaka, K. Minoshima, A. Onae, M. Yasuda, T. Kohnno, S. Kawato, T. Kobayashi, T. Katsuyama, and F.-L. Hong, *Opt. Express* **18**, 1667 (2010).
- N. Kuse, J. Jiang, C. C. Lee, T. R. Schibli, and M. E. Fermann, *Opt. Express* **24**, 3095 (2016).
- C. C. Lee, C. Mohr, J. Bethge, S. Suzuki, M. E. Fermann, I. Hartl, and T. R. Schibli, *Opt. Lett.* **37**, 3084 (2012).
- C. Benko, A. Ruehl, M. J. Martin, K. S. E. Eikema, M. E. Fermann, I. Hartl, and J. Ye, *Opt. Lett.* **37**, 2196 (2012).
- W. Zhang, M. Lours, M. Fischer, R. Holzwarth, G. Santarelli, and Y. L. Coq, *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **59**, 432 (2012).
- T. C. Briles, D. C. Yost, A. Cingöz, J. Ye, and T. R. Schibli, *Opt. Express* **18**, 9739 (2010).
- B. Xu, H. Yasui, Y. Nakajima, Y. Ma, Z. Zhang, and K. Minoshima, *Opt. Express* **25**, 11910 (2017).